ROBERT DYRO

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PROFILE. My expertise is in computational engineering and applied machine learning. My doctoral work in Aeronautics & Astronautics at Stanford University provided me with a deep theoretical understanding and practical experience building systems combining applied machine learning and real-time optimization—work which I published at top statistical and robotics conferences.

As part of the JAX team at Google DeepMind, I focus on core JAX development and large-scale computational engineering for internal research. This includes both high-level distributed computing and low-level kernel development. I have advised and directly worked with production and research teams at Google DeepMind on TPU and GPU performance.

TECHNICAL SKILLS. Programming Languages: *Python, C++*; Scientific Computation: *Julia, Matlab, Maxima*; ML/Deep Learning: *Deep knowledge of the JAX/Pallas kernels/XLA compiler stack on TPUs and GPUs*, several years of experience with *PyTorch on GPUs, Triton, CUDA*; Systems & Tools: *Slurm, GCP.*

CITIZENSHIP. Polish Citizenship (STEM OPT until 2027-06 or another work visa).

EDUCATION

Stanford University — Stanford, CA
PhD, Aeronautics & Astronautics — 2020 - 2024
MS, Aeronautics & Astronautics — 2018 - 2020

University of California, Los Angeles — Los Angeles, CA
BS, Aerospace Engineering, Minor in Philosophy, Summa Cum Laude — 2014 - 2018

WORK HISTORY

GDM Senior Software Engineer, Google DeepMind — 2026 - Present, Mountain View, CA

GDM Software Engineer, Google DeepMind — 2026 - 2026, Mountain View, CA

On the JAX team at Google DeepMind, I focus on core JAX development and large-scale computational engineering for internal research. This includes low-level kernel development, performance optimization for large-scale training and inference, and collaborating on research teams on TPU and GPU performance.

Software Engineer, Google — 2024 - 2026, Mountain View, CA

On the JAX team, I developed high-performance open-source LLM implementations including TPU and GPU examples for Deepseek R1, GPT OSS, Kimi K-2, Qwen 3 MoE and others. I implemented a lightweight serving framework in JAX, achieving full GPU and TPU utilization—matching top commercial implementations. I was one of the few lead kernel performance engineers, working on Pallas TPU kernels for the newest generation of TPU chips (Ironwood). I also consulted with research teams (at e.g., Stanford, Berkeley, Apple) on LLM performance optimizations.

Research Assistant, Autonomous Systems Laboratory, Stanford University — 2020 - 2024, Stanford, CA

My PhD focused on machine learning and optimization in robotics and autonomous vehicle applications. I authored novel real-time optimization and machine learning techniques and focused on consensus optimization, meta-learning (learning through optimization), bilevel programming with differentiable optimization and numerical linear algebra sketching for analysis of machine learning behavior prediction networks.

Research Intern, Cruise — 2022, San Francisco, CA

As a research intern, I worked on Zero-Shot Neural Architecture Search (NAS) internal company research and developed its original production NAS framework.

Research Intern, Toyota Research Institute — 2020, Los Altos, CA

I interned at the self-driving research team and researched human logic embedding—via Signal Temporal Logic (STL)—to computationally capture human-interpretable behavior specification in learned behavior models.

TECHNICAL EXPERIENCE

Relevant Coursework: During my time at Stanford, I took courses focusing on Machine Learning (Meta-learning, Explainable AI, RL, Graph ML, Distribution Shift ML) and Optimization (Convex Optimization, Optimal Control Theory).

My other technical experience includes developing a library for automatic kernel tuning in JAX and Pallas; a differentiable
Pytorch-JAX interoperability layer. My recent experimental projects included a custom quadratic program (QP) solver in CUDA; an experimental tracing-based auto-differentiation library with sparse linear algebra support.

SELECTED PUBLICATIONS

- Dyro, R. (2024). "Optimization-Driven Autonomous Driving: Control, Intent, and Testing." *PhD diss., Stanford University.*
- Dyro, R., Foutter, M., Li, R., Di Lillo, L., Schmerling, E., Zhou, X., and Pavone, M. (2024). "Realistic extreme behavior generation for improved AV testing." *International Conference on Robotics and Automation (ICRA).*
- Dyro, R., Schmerling, E., Arechiga, N., and Pavone, M. (2022). "Second-order sensitivity analysis for bilevel optimization." *International Conference on Artificial Intelligence and Statistics (AISTATS).*
- Dyro, R., Harrison, J., Sharma, A., and Pavone, M. (2021). "Particle MPC for uncertain and learning-based control." *International Conference on Intelligent Robots and Systems (IROS).*

MISC

Philosophy Minor ■ LA Marathon ■ Ham Radio ■ PADI Assistant Instructor ■ I speak English, Polish & German.